Basics Concept of Matrix

What is Matrix?

Matrix: A matrix is a rectangular arrangement of numbers, expressions or symbols in rows and columns enclosed by $(\ \ )$, $[\ \ ]$ or $\|\ \ \|$.

Example: $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ $\quad \begin{bmatrix} a & b & c \end{bmatrix}$ $\quad \begin{bmatrix} a & d & j \\ b & e & h \\ c & f & m \end{bmatrix}$

What do you mean by rows and column?

Row and Column: The row elements are horizontally arranged and column elements are vertically arranged.

Example:

$$\left.\begin{bmatrix} a & d & j \\ b & e & h \\ c & f & m \end{bmatrix}\right\} \longrightarrow \text{Rows}$$

$$\underbrace{\begin{bmatrix} a & d & j \\ b & e & h \\ c & f & m \end{bmatrix}}_{\text{Columns}}$$

What is Dimension of a Matrix?

Dimension: The dimensions, or size, of a matrix are defined as,

$$[\text{Number of Rows} \times \text{Number of Columns}]$$

Example:

$$\begin{bmatrix} a & d & j \\ b & e & h \\ c & f & m \end{bmatrix} \qquad \begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix} \qquad \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$$

3×3 Matrix $\qquad$ 3×2 Matrix $\qquad$ 2×3 Matrix

A matrix is usually denoted by a capital letter and the elements within the matrix are denoted by lower case letters, such as

A matrix [A] with elements a_{ij} is defined by

$$A_{m\times n} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1j} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2j} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{ij} & \cdots & a_{in} \\ \vdots & \vdots & & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mj} & \cdots & a_{mn} \end{bmatrix}$$; i goes from 1 to m and j goes from 1 to n.

Row Matrix: A matrix is said to be a row matrix if it has only one row. A row matrix has only one row but any number of columns.

Example: $\begin{bmatrix} a & b & c \end{bmatrix}_{1\times 3}$ $\qquad$ $\begin{bmatrix} 7 & 3 & 4 & 7 & 0 \end{bmatrix}_{1\times 5}$ $\qquad$ $\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \end{bmatrix}_{1\times n}$

are 3 row matrix with order 1×3, 1×5 and $1 \times n$.

Column Matrix: A matrix is said to be a column matrix if it has only one column. A column matrix has only one column but any number of rows.

Example: $\begin{bmatrix} a \\ b \\ c \end{bmatrix}_{3\times 1}$ $\qquad$ $\begin{bmatrix} 0 \\ 8 \\ 4 \\ 6 \\ 1 \end{bmatrix}_{5\times 1}$ $\qquad$ $\begin{bmatrix} a_{11} \\ a_{21} \\ a_{31} \\ \vdots \\ a_{n1} \end{bmatrix}_{n\times 1}$

are 3 column matrix with order 3×1, 5×1 and $n \times 1$.

Square Matrix: A square matrix has the number of columns equal to the number of rows. A matrix in which the number of rows is equal to the number of columns is said to be a square matrix. Thus an $m \times n$ matrix is said to be a square matrix if $m = n$ and is known as a square matrix of order 'n'.

Example:

$$A = \begin{bmatrix} 1 & 4 & 8 \\ 6 & 2 & 5 \\ 9 & 7 & 3 \end{bmatrix}_{3\times 3}$$

is a square matrix of order 3. In general, $A = [a_{ij}]_{m \times m}$ is a square matrix of order m.

Rectangular Matrix: A matrix where number of rows is not equal to the number of columns is called rectangular matrix.

Example:

$$A = \begin{bmatrix} 1 & 4 & 8 & 3 \\ 6 & 2 & 5 & 5 \\ 9 & 7 & 3 & 7 \end{bmatrix}_{3\times 4}$$

is a matrix of the order 3×4. In general, $A = [a_{ij}]_{m \times n}$ is a square matrix of order $m \times n$.

Problem: Find the dimension of each matrix with type.

1. $A = \begin{bmatrix} 2 & -1 \\ 0 & 5 \\ -4 & 8 \end{bmatrix}$

4. $A = \begin{bmatrix} 2 & 5 & 3 & 8 \\ 0 & 6 & 7 & 1 \\ -4 & 7 & 4 & 0 \end{bmatrix}$

7. $C = \begin{bmatrix} 0 \end{bmatrix}$

2. $B = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$

5. $A = \begin{bmatrix} 3 & 1 & 6 & 5 \\ 2 & 6 & 4 & 5 \\ 7 & 9 & 8 & 6 \\ 2 & 8 & 0 & 9 \end{bmatrix}$

8. $C = \begin{bmatrix} 2 & 3 & 2 & 0 & 6 \end{bmatrix}$

3. $C = \begin{bmatrix} 0 & 5 & 3 & -1 \\ -2 & 0 & 9 & 6 \end{bmatrix}$

6. $C = \begin{bmatrix} 2 & 0 \end{bmatrix}$

9. $C = \begin{bmatrix} 1 & 3 \\ 2 & 8 \end{bmatrix}$